24-780 Final Project Conceptual Design
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Product Name: Robot Arm Air-Hockey

Note*: These are initial design concepts and may be subject to change during further conversations and findings.*

Project Description
This project is a 2D top-view air hockey game in which a human player competes against a computer-controlled 2 link robotic arm (or mobile robot). The arm is mounted centered and behind its own goal and strikes the puck with a paddle at its tip. Its joints have a maximum rotation speed so that the arm cannot always reach the puck in time. This speed limit is the main control for difficulty because a slower arm is easier to beat and a faster one can reach more shots and recover more quickly.

The arm operates in both attacking and defending modes and defaults to defending. The defending mode is active when the puck is moving faster than a threshold speed and the arm predicts where the puck will cross an imaginary defense line in front of the goal. It then moves the paddle there to intercept it. This speed threshold can also increase with difficulty so that the arm becomes more aggressive in harder game modes. When the puck's speed is below the threshold, the arm switches to an attacking mode and chooses a point along the puck's path to intercept it at. From that strike point, it calculates candidate shots toward 5 equally spaced points on the opening of the player's goal, using 0, 1, or 2 wall deflections. It discards any paths that pass within a set margin of the player's paddle at that moment and then selects the remaining path that requires the least movement of the arm. If there are no shots the arm returns to defending. The mobile robot, a wheeled robot carrying a mallet on its half of the table, follows the same decision logic. Instead of joint speed, its difficulty is set by its wheel speed and acceleration limits.

Product Identification
The product is a 2D top-view air hockey game in which a human player competes against a robot opponent. The puck slides across the table with friction, bounces off the side walls, and scores when it enters a goal. It reacts to mallet hits based on the mallet's position and velocity. The player controls a mallet with the mouse, and the mallet stays on the player's half of the table.

On the title screen the player chooses between two opponent modes. In arm mode the opponent is a 2-link robotic arm mounted at the robot's goal. In mobile robot mode it is a wheeled robot carrying a mallet on the robot's half of the table. In both modes the robot defends its own goal and attacks the player's goal. The player also chooses a difficulty. It is set by the robot's speed limits (joint speed for the arm, wheel speed and acceleration for the mobile robot) and by how aggressively the robot plays.

The game follows a complete flow. On the title screen the player chooses the opponent and difficulty, then starts a game or quits. During play the score is shown on screen, sound effects accompany the action, and the game pauses briefly after each goal. When the game ends, a game-over screen appears and the program returns to the title screen.

What are the special features of the product?
The main feature is a physically limited opponent. The robot is not a mallet that teleports to the right spot. The arm is driven through forward and inverse kinematics, with limits on joint angles and joint speeds. The mobile robot is driven by its wheel speeds, with limits on speed and acceleration. Before committing to a move, both robots estimate whether they can reach the point in time. As a result, the robot can be beaten by shots that are hard for its mechanism to reach.

The robot's decisions follow two modes. When the puck is faster than a threshold, the robot defends: it predicts the puck's path, including wall bounces, and intercepts the puck at an imaginary defense line in front of its goal. When the puck is slower than the threshold, the robot attacks. It chooses an intercept point along the puck's path and generates 25 candidate shots: shots with 0, 1 or 2 wall deflections, each aimed at 5 equally spaced points across the player's goal. It then picks the shot that needs the least robot movement, meaning the least joint motion for the arm or the shortest driving distance for the mobile robot.

What are the performance targets?
The game will offer at least three difficulty levels: Easy, Medium and Hard. Switching between the two opponent modes on the title screen will need no change to the opponent logic, because both robots use the same interface. For shot planning, all 25 candidate shots will be generated and scored in under 1 ms, so the robot can re-plan every frame.

Market Identification
Our target users are gamers who want a quick and easy game they can play on a computer. The controls are simple, so no gaming experience is needed. Air hockey is a well-known game that many people have played, and there are many free air hockey games on phones and computers, which shows that there is steady interest. Some competing products are mobile, browser air hockey games, and real air hockey tables at arcades (e.g. Air Hockey Gold on the App Store, and many free web games). They are easy to play and often have difficulty levels and 2-player mode. However, in most of them, the computer is often just a mallet that moves freely on the table, and difficulty is often just a change in speed or reaction. Our product is different in the sense that in our game, the robot opponent has real physical limits. The robot arm has joint angle and speed limits, and the mobile robot has wheel speed and acceleration limits. So the robot cannot always reach the puck in time, and players can win by making shots that are hard for the robot to reach. Players can also choose between two types of robots and different difficulty levels. It is free to run on a computer and needs no extra hardware.

System Description

- **Shared Core (Everyone)**

        - Common math types (2D vectors) and table dimensions/constants
        - Shared game state (puck, mallets, scores)
        - A common robot interface, so the two robots can be swapped freely
    - **Table & Puck Physics (Noah)**
        - Puck motion with friction
        - Wall bounces and goal detection
        - Mallet–puck collisions, using mallet position and velocity
        - Human mallet movement (follows the mouse, kept on the player's half)
        - Predicts the puck's future path for the opponent logic
    - **Robot Arm (Shayan)**
        - 2-link arm mounted at the robot's goal
        - Forward and inverse kinematics
        - Joint angle and speed limits (speed sets difficulty)
        - Estimates time to reach a point; carries out moves and strikes
        - Draws the arm
    - **Mobile Robot (Harrison)**
        - Wheeled robot carrying a mallet on the robot's half of the table
        - Converts desired motion into wheel speeds
        - Wheel speed and acceleration limits (sets difficulty)
        - Estimates time to reach a point; carries out moves and strikes
        - Draws the robot
    - **Opponent Logic (Anthony)**
        - Decision-making that works with either robot
        - Modes: guard, defend, attack, recover
        - Defense: predicts the puck's path and intercepts it in time
        - Attack: generates direct and wall-bank shots, then picks the best one
    - **Game Flow (Everyone)**
        - Title screen: choose opponent and difficulty, start, or quit
        - Gameplay loop and timing
        - Scoring, goal pause, game over, return to the title screen
        - On-screen score display and sound effects

All components share a common core that defines the table, game state, and a single robot interface. The opponent logic sends commands through this interface, so the same decision-making controls either the arm or the mobile robot, and the physics only needs each mallet's position and velocity. Game Flow runs the main loop, calling each component in order of every update (input, opponent decision, robot motion, puck physics, then scoring and drawing).